

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	11 February 2026
Team ID	LTVIP2026TMIDS82993
Project Name	HouseHunt: Finding Your Perfect Rental Home
Maximum Marks	4 Marks

Technical Architecture:

Architecture Type: 3-Tier MERN Stack Architecture

The application follows a 3-Tier Architecture:

1 Presentation Layer (Frontend)

- React.js Web Application
- Role-based UI (Admin / Owner / Renter)
- REST API communication using Axios

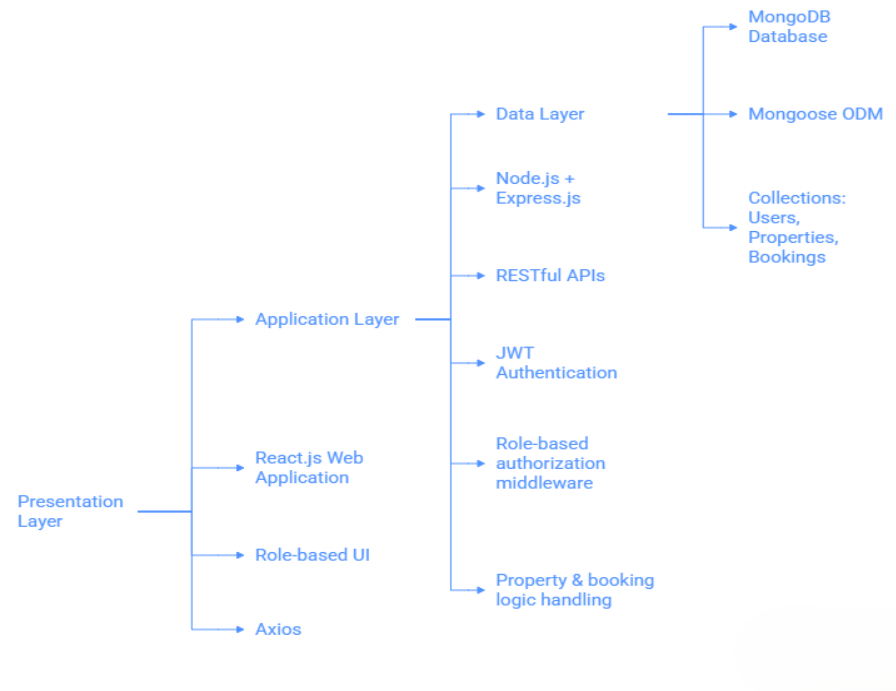
2 Application Layer (Backend)

- Node.js + Express.js
- RESTful APIs
- JWT Authentication
- Role-based authorization middleware
- Property & booking logic handling

3 Data Layer

- MongoDB Database
- Mongoose ODM

3-Tier MERN Stack Architecture



- Collections: Users, Properties, Bookings

 Infrastructure Demarcation

Component	Location
Frontend	Local Development (React Dev Server)
Backend	Local Node.js Server
Database	Local MongoDB Instance

 External Interfaces

- REST API Communication between Frontend & Backend
- JWT Token Authentication
- MongoDB Database Connection

Table-1: Components & Technologies:

S.No	Component		Description
1.	User Interface		Web-based UI for Admin, Owner, and Renter
2.	Application Logic-1		User authentication and authorization
3.	Application Logic-2		Property management (CRUD operations)
4.	Application Logic-3		Booking and dashboard logic
5.	Database		Stores user, property & booking data

6.	File Storage		Property image storage
7.	External API-1		REST API communication
8.	External API-2		(Future) Payment API Integration
9.	Machine Learning Model		Not Applicable
10.	Infrastructure (Server / Cloud)		Local Server Deployment

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	MERN Stack Frameworks	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	JWT Authentication, Password Hashing	JWT, bcrypt.js
3.	Scalable Architecture	3-Tier Architecture (Frontend–Backend–Database)	REST API Architecture
4.	Availability	Runs locally; scalable to cloud deployment	Node.js Server
5.	Performance	Lightweight REST APIs, Asynchronous Handling	Express.js, Axios